

MIHAIL KOGALNICEANU, Romania

WMO: 154810

Lat: 44.362N

Lon: 28.488E

Elev: 108

StdP: 100.04

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.1	-8.8	-13.2	1.2	-9.9	-11.0	1.5	-7.4	15.1	-2.7	13.6	-2.7	5.6	0	0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	33.1	21.1	31.8	20.9	30.1	20.6	23.9	28.3	23.0	27.7	22.1	26.9	4.9	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	17.9	25.6	21.8	16.7	24.8	20.8	15.7	24.1	72.3	28.8	68.6	27.4	65.3	26.4	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.6	8.5	DB	-14.9	35.9	2.5	1.9	-16.7	37.2	-18.1	38.3	-19.5	39.3	-21.3	40.7	(4)
(5)			WB	-15.2	25.6	2.5	1.3	-17.0	26.5	-18.4	27.3	-19.8	28.0	-21.6	28.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	0.2	2.2	6.2	10.9	16.5	21.0	23.6	23.5	18.6	12.7	7.5	1.9
	DBStd	9.10	5.02	5.14	4.21	3.57	3.29	2.88	2.47	2.83	3.32	4.13	4.71	4.68
	HDD10.0	1064	304	220	131	29	1	0	0	0	0	21	103	254
	HDD18.3	2749	562	451	376	224	75	9	1	1	37	177	326	511
	CDD10.0	1839	0	2	13	57	203	331	422	417	259	106	27	2
	CDD18.3	484	0	0	0	1	18	90	164	161	46	4	0	0
	CDH23.3	4374	0	0	2	21	164	792	1534	1503	326	33	1	0
	CDH26.7	1543	0	0	0	4	26	246	589	603	72	3	0	0
Wind	WSAvg	4.2	4.8	4.8	4.8	4.4	3.9	3.8	3.8	4.1	4.1	4.2	4.6	
Precipitation	PrecAvg	478	31	21	33	35	50	55	51	37	52	39	34	38
	PrecMax	635	74	61	77	99	128	129	135	126	115	77	70	92
	PrecMin	256	3	2	1	3	12	16	8	5	6	3	1	3
	PrecStd	101	21	16	19	21	30	27	34	33	36	20	20	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	18.0	21.9	26.0	29.1	33.0	36.0	36.0	31.8	26.9	21.9	15.2
		MCWB	10.5	11.8	12.4	15.7	18.6	21.9	20.7	21.3	19.6	18.0	15.9	12.3
	2%	DB	10.2	14.1	18.8	22.1	27.0	31.0	33.1	33.8	28.8	23.8	18.1	12.2
		MCWB	8.3	10.5	11.4	14.7	18.5	20.6	21.1	20.8	18.8	17.0	14.7	10.2
	5%	DB	8.1	11.1	15.8	19.8	25.1	29.8	31.2	31.9	26.9	21.1	15.8	10.1
		MCWB	7.3	8.6	10.4	13.9	17.5	20.5	21.1	20.9	18.7	16.3	13.7	9.2
	10%	DB	6.2	8.2	13.0	17.2	23.2	27.9	30.0	30.0	25.1	19.0	13.9	8.1
		MCWB	5.7	6.9	8.9	12.3	16.7	19.7	20.9	20.6	18.4	15.1	12.3	7.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	13.3	14.0	18.0	21.4	24.0	25.1	26.1	22.9	20.4	17.0	12.8
		MCDB	13.1	16.0	19.5	23.3	25.5	28.5	28.8	30.0	25.7	23.3	19.5	14.6
	2%	WB	8.7	10.9	12.2	15.8	20.0	22.7	23.8	24.2	21.5	18.5	15.4	10.8
		MCDB	10.0	13.7	17.2	19.8	24.5	27.5	28.3	28.5	24.9	21.4	17.8	12.2
	5%	WB	7.1	8.9	10.7	14.5	18.7	21.7	22.9	23.0	20.4	17.2	14.0	9.2
		MCDB	7.8	10.9	14.9	18.7	23.2	26.5	27.8	27.8	23.9	20.0	15.5	10.1
	10%	WB	5.8	7.1	9.3	13.1	17.6	20.8	22.1	21.9	19.4	15.9	12.6	7.4
		MCDB	6.3	8.2	12.4	16.5	21.7	25.9	27.2	26.8	23.2	18.6	13.8	8.0
Mean Daily Temperature Range	5% DB	MDBR	6.3	7.7	9.5	10.2	10.7	11.1	11.3	11.4	10.5	9.2	7.4	6.7
		MCDBR	9.3	12.2	14.3	13.9	13.3	13.3	13.1	13.5	12.8	12.2	10.6	9.3
		MCWBR	7.3	8.5	8.4	7.9	6.6	6.0	5.5	5.8	5.9	6.8	7.2	7.5
	5% WB	MCDBR	8.7	11.4	12.9	12.1	11.8	11.6	11.2	11.6	10.7	10.8	9.4	9.0
		MCWBR	7.2	8.5	8.2	7.8	6.8	6.1	5.7	6.0	6.2	7.1	6.9	7.5
Clear-Sky Solar Irradiance	taub	0.345	0.373	0.404	0.439	0.416	0.418	0.431	0.441	0.416	0.387	0.364	0.345	
	taud	2.320	2.252	2.202	2.126	2.275	2.301	2.275	2.239	2.281	2.333	2.339	2.338	
	Ebn at Noon	754	797	823	824	856	854	837	814	802	774	725	713	
	Edh at Noon	83	105	126	147	131	128	130	130	115	96	81	74	
All-Sky Solar Radiation	RadAvg	1.30	2.10	3.32	4.78	6.04	6.57	6.57	5.88	4.25	2.67	1.51	1.15	
	RadStd	0.19	0.28	0.34	0.47	0.37	0.37	0.34	0.32	0.32	0.25	0.19	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		+0.49	N/A	N/A	N/A	N/A	-0.77	N/A	N/A	+115	+74	(46)	
(47) Regional (31 neighbors)		+0.49	N/A	N/A	N/A	N/A	N/A	N/A	-122	+89	+57	(47)	

Nomenclature: See separate page